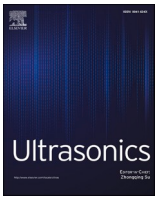
Ultrasonics 132 (2023) 107026



Contents lists available at ScienceDirect

Ultrasonics

journal homepage: www.elsevier.com/locate/ultras





Physics-informed neural networks for transcranial ultrasound wave propagation

Linfeng Wang [a], Hao Wang [a], Lin Liang [b], Jian Li [a], Zhoumo Zeng [a], Yang Liu [a,*]

[a] State Key Laboratory of Precision Measuring Technology and Instruments, Tianjin University, Tianjin 300072, China
[b] Schlumberger-Doll Research, One Hampshire St, Cambridge, MA 02139, USA

A B S T R A C T

Transcranial ultrasound imaging has been playing an increasingly important role in the non-invasive treatment of brain disorders. However, the conventional mesh-based numerical wave solvers, which are an integral part of imaging algorithms, suffer from limitations such as high computational cost and discretization error in predicting the wavefield passing through the skull. In this paper, we explore the use of physics-informed neural networks (PINNs) for predicting the transcranial ultrasound wave propagation. The wave equation, two sets of time snapshots data and a boundary condition (BC) are embedded as physical constraints in the loss function during training. The proposed approach has been validated by solving the two-dimensional (2D) acoustic wave equation under three increasingly complex spatially varying velocity models. Our cases demonstrate that due to the meshless nature of PINNs, they can be flexibly applied to different wave equations and types of BCs. By adding physics constraints to the loss function, PINNs can predict wavefields far outside the training data, providing ideas for improving the generalization capability of existing deep learning methods. The proposed approach offers exciting perspectives because of the powerful framework and simple implementation. We conclude with a summary of the strengths, limitations and further research directions of this work.

1. Introduction

As an emerging diagnostic technique, transcranial ultrasound imaging has drawn increasing attention in the fields of stroke, cerebrovascular disease [1], and even Alzheimer's [2] and Parkinson's disease [3]. It has become an attractive health care solution for being real-time, non-invasive, low cost and highly portable compared with other imaging modalities [4]. The use of computational ultrasound models has the potential to predict the ultrasound field as it propagates through the skull in the brain, thus explaining object-specific dose and targeting changes [5]. Therefore, the ability to simulate the propagation of ultrasound through human body is critical to the approval and success of clinical applications. The wave equation is one of the most widely studied hyperbolic partial differential equations (PDEs) in scientific and medical applications because of its excellent illustrative capabilities [6]. An effective and accurate approach to transcranial ultrasound modeling can significantly reduce the time and money costs for patients. It also plays a major role in the design and customization of medical ultrasound transducers [7]. Therefore, it is important to accurately model the transcranial ultrasound propagation by solving the PDEs.

In the literature, PDEs are commonly solved by traditional methods such as finite element method (FEM), finite volume method (FVM), and finite difference method (FDM), which have been successfully applied in nondestructive testing [8,9,49,50], medical imaging [10], metal materials [11], geophysical [12] and other fields. However, these mesh-based methods suffer from the same limitations. They share the same computational concept and require a grid setup before solving PDEs. When the grid is coarsely divided, the accuracy of the solution is undoubtedly reduced. When the grid is very small to ensure the accuracy, it will greatly increase the occupation of computational resources. As the dimensionality of the problem increases, the well-known curse of dimensionality problem arises. What's worse, it becomes more difficult for these mesh-based PDEs solvers to deal well with domains having irregular topography because of the poor quality of meshing [13]. These are considered from the perspective of spatial meshing. In addition, the errors caused by discretization based on temporal meshing cannot be ignored [14]. The PDEs are solved through time-stepping in these methods. In order to meet the resolution of the system, the size of the temporal grid needs to be controlled to improve the time sampling. Given the constraints of computing resources, there is no guarantee that this is always possible for sufficiently small spatial and temporal grid. It is not always easy to find a balance between solution accuracy and computational cost. Therefore, it is of great significance to find an alternative method to obtain the solution of PDEs.

In the past few years, numerical simulation work for ultrasound wave propagation has never stopped developing. This has gradually led to a

* Corresponding author.
E-mail address: ultrasonicslab@tju.edu.cn (Y. Liu).

https://doi.org/10.1016/j.ultras.2023.107026
Received 30 October 2022; Received in revised form 9 April 2023; Accepted 24 April 2023
Available online 28 April 2023
0041-624X/© 2023 Elsevier B.V. All rights reserved.

deeper understanding of propagation mechanisms and advances in imaging techniques in various fields including medical applications. Along with recent advancements, neural networks (NNs) have emerged as a new tool for dealing with PDEs. Solving PDEs by NNs offers several advantages. For example, the solutions thus obtained are continuous and differentiable, so that we can easily get other physical parameters by performing integral or differential calculations. Solutions with high applicability over the entire domain can be obtained accurately with a small number of training points. In addition, a relatively simple modification of the network can cope with different boundary conditions (BCs) of the equations [15]. Hughes et al. [16] designed a recurrent neural network (RNN) structure based on the physical wave principle that can perform vowel recognition according to the information of different vowels after passing through an inhomogeneous medium. Moseley et al. [17] presented two types of NNs that can simulate the propagation of seismic waves in horizontally layered and faulted two-dimensional (2D) acoustic media, using WaveNet and convolutional autoencoder as frameworks, respectively. The convergence of the networks was improved by introducing a time-dependent gain term in the loss function. Both networks are even hundreds of times faster than FDM. Gantala et al. [18] proposed a Data driven-simulation-assisted-Physics learned AI (DPAI) model based on convolutional long short-term memory (ConvLSTM) and an encoder-decoder. The DPAI was trained using a dataset generated by FEM and successfully simulated the propagation of transient ultrasonic waves in a 2D domain. Ahmed et al. [19] also proposed a DSWE network using ConvLSTM and a convolutional encoder to construct Young's modulus maps based on the velocity data generated by a single acoustic radiation force push. However, most of the data-driven NNs rely on establishing nonlinear relationships between the inputs and outputs. They often require a large amount of labeled data for training. Because of this, these methods are relatively dependent on the training set, and it is a challenge to simulate accurately for those outside of the training distribution [17]. Stanziola et al. [20] developed a UNet-based fast iterative solver of the 2D heterogeneous Helmholtz equation with a learnable hidden state for modeling the transcranial wavefield. The trained solver not only performs well on the test set, but more importantly, still generalizes well beyond the training set for much larger solution domains, more complex excitation sources and velocity distributions. However, this method still needs to establish a large training set of data, which is time-consuming. Moreover, this method cannot obtain the time domain signal of the wavefield at arbitrary times.

Physics-informed neural networks (PINNs) [21], one of the latest achievements in recent years, is an approach that essentially embeds physical constraints into the loss function as an alternative to purely data-driven traditional NNs [22]. Furthermore, the automatic differentiation technology in the deep learning software package is also an important factor for the rapid development of PINNs. This technology makes it very convenient to obtain the gradient of network parameters and further differentiate the network weight to train the network [23]. In PINNs, automatic differentiation technology is mainly used on the differentiation of the NN's output to the input, and the result is taken as a part of the loss function. Besides, the errors in terms of the initial and boundary conditions (ICs/BCs) are also components of the loss function. These error terms together constrain the training of the network to solve the equation. An obvious advantage of this meshless approach is that it is easy to implement and can be simply extended to arbitrarily complex computational domains and high-dimensional PDEs, which is a challenge for conventional mesh-based approaches [24]. The rise of PINNs provides new ideas for the development of a large number of follow-up work. Song et al. [13] used PINNs to obtain the scattered pressure wavefields of a transversely isotropic (TI) medium with a vertical axis of symmetry (VTI) in a very short time and validated the method on 2D and three-dimensional (3D) inhomogeneous models with irregular topography. Karimpouli et al. [25] obtained solutions of the one-dimensional (1D) time-dependent seismic wave equation with high accuracy from a small amount of data, and proved that any other parameters involved in the equation can also be gained by PINNs. Mao et al. [26] used PINNs to solve both the forward and inverse problems of the Euler equation for high-speed aerodynamic flows in 1D and 2D domains, and illustrated the superiority of PINNs in certain specific inverse problems. Rautela et al. [27] presented a physical knowledge assisted machine learning method (PKaML) and achieved damage detection and location of a carbon fiber-reinforced polymer plate. PKaML showed better performance than the direct deep learning method in terms of training time and accuracy in a random noise environment. More importantly, it can reduce the number of sensors required in actual applications. Van et al. [28] trained PINNs to fit the perfusion magnetic resonance (MR) data under the physical law of multi-compartment exchange model, and validated the method *in silico* and *in vivo*. Chen et al. [29] introduced the PINN-SR method (PINNs with sparse regression), which can predict the key parameters in the governing equations of nonlinear systems from sparse and noisy data. Raissi et al. [30] developed a hidden fluid mechanics (HFM) based on the framework of PINNs, which addressed the problem of flow visualization by embedding the Navier-Stokes equation into the networks. Lu et al. [31] solved different types of equations using DeepXDE and improved the training efficiency of PINNs by applying a new residual-based adaptive refinement (RAR) method. PINNs have shown excellent performance in many forward and inverse problems in the fields of science and engineering. However, few studies have been conducted regarding their use with transcranial ultrasound wave propagation.

In this paper, we first propose an approach for transcranial ultrasound wave propagation using PINNs. We extend the original PINNs method to predict the wavefield of transcranial ultrasound at any moment and any position, requiring two early-time snapshots as training data. In this study, the snapshots in terms of displacement are obtained by the spectral-element method (SEM). It is worth noting that conventional PDEs solvers cannot simulate transcranial ultrasound propagation while using only two early-time wavefield snapshots, which highlights the superiority and flexibility of PINNs in solving the wave equation with a small amount of data. Furthermore, the numerical solutions obtained by SEM are considered as the reference solutions to evaluate the effectiveness and accuracy of the prediction results of PINNs. The performance of the proposed method has been verified on a homogeneous model, an inhomogeneous model and a 2D brain model with and without a clot, respectively. For the first two cases, it has been proved that the proposed method has impressive generalization capability and can match the wavefield solutions well given only two early-time snapshots. For the brain model, PINNs can also predict most information including reflected wave when propagates to the skull. Although there is a loss of information of the predictions inside the brain, the time domain signals in the $\times$ direction and y direction at the receiving sensors indicate that these losses are acceptable. Besides, it is not difficult to apply the algorithm to any complex computational boundary because of the meshless nature of PINNs. The following contents of this paper are as follows: a brief overview of the methodology for transcranial ultrasound wave propagation and PINNs is described in Section II. Section III carries out numerical simulations under different conditions and the results are illustrated. The advantages and limitations of the proposed method are discussed in Section IV and final conclusions are summarized in Section V.

2. Methodology

2.1. 2D acoustic transcranial ultrasound wave equation

In general, the wave propagation of transcranial ultrasound can be very complex due to many issues such as inhomogeneous velocity and density distributions, shear wave effects in the skull, nonlinear effects, and different acoustic absorption coefficients involved [32,33]. However, when the ultrasound is propagating through the skull close to the

normal incidence, the shear wave effects can be ignored [34,35]. In addition, the nonlinear effect is mainly used in ablation treatment, and is only effective in a small range near the lesion [36]. Therefore, this process can be described by the following linear acoustic wave equation in an inhomogeneous absorbing medium after simplification [35,37]:

$$\left(1+\tau_0(r)\frac{\partial}{\partial t}\right)\left[\rho_0(r)\nabla\cdot\left(\frac{1}{\rho_0(r)}\right)\nabla p(r,t)\right]-\frac{1}{c_0(r)^2}\frac{\partial^2 p(r,t)}{\partial t^2}=S_0(r,t), \quad (1)$$

where τ_0 represents the spatial distribution of the absorption coefficient, r is a general space coordinate, ρ_0 stands for the spatial distribution of the density, p is the pressure in the acoustic medium, c_0 and S_0 represent the distribution of the wave velocity and the acoustic source. The above equation accurately describes the law of wave propagation when the velocity, density and absorption coefficient are inhomogeneously distributed. If the acoustic absorption is treated as a second-order effect [38], Eq.(1) describing the wave propagation of transcranial ultrasound can be further simplified as [20]:

$$\rho_0(r)\nabla\cdot\left(\frac{1}{\rho_0(r)}\right)\nabla p(r,t)-\frac{1}{c_0(r)^2}\frac{\partial^2 p(r,t)}{\partial t^2}=S_0(r,t). \quad (2)$$

Eq.(2) can be expressed as Eq.(3) with the scalar wave potential ϕ:

$$\nabla^2\phi-\frac{1}{c_0^2}\frac{\partial^2\phi}{\partial t^2}=S_0, \quad (3)$$

where $\nabla^2=\frac{\partial^2}{\partial x^2}+\frac{\partial^2}{\partial y^2}$ is the Laplace operator in the 2D Cartesian coordinate system.

Although the wave equation is linear, it is well known that due to the oscillatory, broadband, and dispersion of the solution, it is never easy to solve it in complex medium [22]. The physical laws involved in this wavefield can also be very complex, requiring consideration of phenomena including reflection, refraction, transmission and scattering at the junction of the different velocity distributions or the boundary of the domain, the simultaneous interaction of multiple types of waves, and the corresponding change in amplitude as the wave propagates [39]. The fact that the skull is solid–liquid coupled, multi-layered and porous, coupled with the nature of the velocity of the skull being much faster than the internal tissues, increases the difficulty of modeling the propagation of transcranial ultrasound. Therefore, the problem may be even more difficult to solve than some nonlinear problems that have been addressed with PINNs, such as fluid mechanics.

2.2. The principle of PINNs

It has been proved that the Feed-forward neural network (FNN) is sufficient to solve PDEs [15]. This is one of the reasons why the architecture of PINN is almost completely based on a fully connected FNN in the current literature. In this paper, we define a fully connected FNN with three channels of inputs: the spatial coordinates x, y and time t. The output of the network is the predicted value of the scalar wave potential ϕ and the network has L hidden layers. The displacement variable in the $\times$ and y directions (u_x and u_y) during wave propagation can be derived by automatic differentiation of ϕ:

$$u_x=\frac{\partial\phi}{\partial x} \quad (4)$$

$$u_y=\frac{\partial\phi}{\partial y}$$

In FNN, the value of the current layer is calculated from the previous one as:

$$q^i=\sigma^i\left(\omega^i x^i+b^i\right), i=1,2,\cdots,L \quad (5)$$

where σ represents the activation function, such as sine function, hyperbolic tangent activation function and sigmoid function, and q stands for the values of ith layer. The training of FNN is essentially an optimization process. In each iteration, the weights ω and biases b of the network are updated by minimizing the loss function. Mean Square Error (MSE) is one of the most commonly used loss functions, which is defined as follows:

$$MSE=\frac{1}{N_u}\sum_{i=1}^{N_u}(u-u^*)^2, \quad (6)$$

where u and u^* are the reference solution and the predicted value output by the network and N_u stands for the number of points selected to calculate the *MSE*. Then we sample a certain number of training points from two early-time snapshots and boundary data to train FNN, making it satisfy Eq.(3) at the same time. In fact, the outputs of the network must meet the PDE, the two snapshots, and BC to achieve the best results. Therefore, the training process can be transformed into minimizing the weighted sum of the *MSEs*:

$$MSE(\Theta)=\lambda_{PDE}MSE_{PDE}+\lambda_{S1}MSE_{S1}+\lambda_{S2}MSE_{S2}+\lambda_{BC}MSE_{BC}, \quad (7)$$

where $\Theta=\omega\bigcup b$ stands for the union of all the weights and biases in the network. MSE_{PDE}, MSE_{S1}, MSE_{S2}, and MSE_{BC} represent the PDE error function, the first snapshot error function, the second snapshot error function and BC error function using in Eq.(6) respectively. In this paper, the IC is regarded as a kind of BC. λ_{PDE}, λ_{S1}, λ_{S2}, and λ_{BC} are the weights of the error functions that are adjusted according to different situations to converge to the correct solution. Improper weighting of loss functions will slow down the convergence speed of the network training, and even affect the prediction accuracy seriously. Compared with the conventional PDEs solvers, the proposed method predicts the solution by solving an optimization problem which is carried out by updating the weight and biases from each iteration to minimize Eq.(7), i.e.,

$$\Theta^*=argmin\{MSE(\Theta^*)\}. \quad (8)$$

We use Adam as the optimization function [40] with a suitably chosen batch size (BSGD) for all the cases in this paper:

$$(\omega_{next},b_{next})=(\omega_{current},b_{current})-\xi\frac{\partial MSE}{\partial(\omega_{current},b_{current})}, \quad (9)$$

where ξ is the learning rate during training. Training stops when the network iterations reach saturation or when the training epoch reaches a set value.

Unlike traditional numerical methods, since the solutions of PINNs are obtained by solving a non-convex optimization problem, there is no guarantee of unique solutions for PINNs. In practice, we need to tune all the hyperparameters to achieve a satisfactory level of accuracy. The required network size depends largely on the smoothness of the solution of PDEs. The setting of hyperparameters in the network cannot be neglected, which will affect the convergence of PINNs to different solutions [31,41]. Therefore, we need to train PINNs for a few times with different hyperparameters and select the network with the least training loss as the final solution. More importantly, a normalization step is required before training. It consists of mapping the input and output in the interval $[-1,1]\in\mathbb{R}$ and scaling Eq.(3) to constrain the wave speed in the interval $[0,1]\in\mathbb{R}$.

2.3. PINNs design and workflow

Fig. 1 shows the workflow of PINNs for transcranial ultrasound wave propagation. The input of the network is the spatial coordinates x, y and time t, and the output is the prediction of the wavefield for this coordinate at this moment. According to the complexity of the medium, we adjust the number of hidden layers and neurons of the network to achieve the best convergence result. We use the identity function as the activation function between the last hidden layer and the output, and $\sigma=tanh(\bullet)$ function is chosen for the rest of the hidden layers. In this paper the early-time snapshots in terms of displacement used for

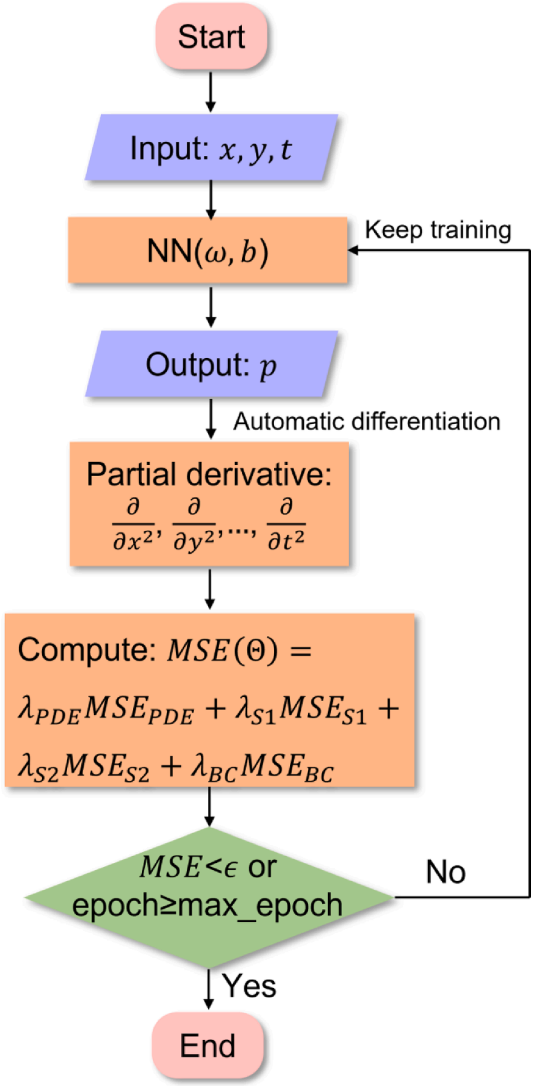


Fig. 1. Schematics of PINNs' workflow for transcranial ultrasound wave propagation. We think the training stops when *MSE* becomes smaller than the threshold (ϵ), or the training reaches the set value (max_epoch).

training are obtained by the SEM. After setting the parameters such as velocity model, density model, location and frequency of excitation source as ICs, we carry out forward simulation with a few time steps to obtain the reference solutions of early wavefields. Importantly, only two early-time snapshots are used to train the network. After training, we use the wavefield at later time steps from forward simulation to test the generalization ability of PINNs. For each update step, a set of points are sampled from the training data to calculate the *MSEs* in Eq.(7). A batch size of 2000 is set for each update step with a learning rate ξ of 3×10^{-4} in all cases.

3. Results

We used three increasingly complex velocity models in Fig. 2 to evaluate the predictive ability of the proposed method for the propagation of transcranial ultrasound. It is necessary to reset the parameters of FNN and the weights in Eq.(7) to retrain PINNs for each velocity model. The models include: (1) a homogeneous model with a fixed sound velocity of 1500 m/s; (2) an inhomogeneous model including 1500 m/s and 2000 m/s and (3) a 2D brain model, the outer part of which is considered to be surrounded by water with a velocity of 1500 m/s [42]. The multimodal imaging-based detailed anatomical (MIDA) model is a segmented cranial phantom which has been created from a combination of scans taken with magnetic resonance imaging, magnetic resonance angiography, and diffusion tensor imaging [43]. We used the the human head data in MIDA model as the original data sample. 465 slices were obtained with aninterval of 0.5 mm, and one of them was selected to build the 2D brain model *insilico* simulations. The data in MIDA model is separated according to tissue types, and each has its unique physical properties. The data also combines the sound velocity and density of different tissue types from [33–37,44]. However, the acoustic characteristics of small tissues may be difficult to measure in practice (especially *invivo*), resulting in little or no recording of the acoustic characteristic parameters of these tissues. In these cases, we use parameters from other tissues similar in structure and composition to the tissue for estimation, provided that such estimation will not affect the wave propagation. Fig. 2(d) shows the brain slice selectedfrom MIDA, which can be used for generating reference solutions and PINNs training after being transformedinto the 2D brain velocity model (Fig. 2(c)).

3.1. Homogeneous model

In this case, we explore the ability of PINNs under a homogeneous model as shown in Fig. 2(a). We choose a Gaussian point source with a dominant frequency of 20 kHz at (0.5,0.8) as the excitation source. There are 7 hidden layers in FNN and each contains 40 neurons. In the process of propagation with a constant velocity, the amplitude of the wave will gradually decrease due to the spherical dispersion. In this case, all the surfaces are absorbing boundaries except for free-surface condition for the top of the domain. For MSE_{PDE} term, we randomly sample 10,000 points for training, while for two early-time snapshots, we sample 6400 points respectively. Another 5000 points are needed to train MSE_{BC} terms, due to the free-surface condition for the top surface. The weight values for different *MSE* terms in Eq.(7) are $\lambda_{PDE} = 0.1$, $\lambda_{S1} = \lambda_{S2} = 1$, and $\lambda_{BC} = 0.01$. When the loss items are almost no longer reduced or have reached the arranged value of epoch, we consider the network training to be over. The time step is set to $5 \times 10^{-8} s$, and we simulate 0.5 *ms* for wave propagation. Fig. 3(a1) to (a4) show the reference solutions obtained by the SEM at 0.10 *ms*, 0.12 *ms*, 0.33 *ms* and 0.45 *ms*, respectively. Among them, only the first two early-time snapshots are used for training. Fig. 3(b1) to (b4) represent the predictions of PINN at corresponding times. Fig. 3(c1) to (c4) represent the errors between them. The prediction results agree with the reference solutions accurately, capturing both its kinematics and amplitude attenuation. It is important to highlight that no other labeled data were used throughout the training process except for two early-time snapshots that contained information about the excitation source. After setting the BC loss term, PINN successfully applies the BCs to the corresponding surfaces, capturing the reflected waves from the top surface as shown in Fig. 3(b3) and (b4). Although the training data does not contain information about the reflections, PINN can still accurately predict the position and amplitude of the reflected waves with the maximum error being less than 3%. In addition, since the solution of PINN is smooth and infinitely differentiable, it can automatically satisfy the absorbing BC, as shown in Fig. 3(b4). We can effectively choose different BCs for PINN by setting the BC loss term, which is exactly what we want to demonstrate through this case. After offline training, we can obtain the time domain signal of the point by inputting the space coordinates *x* and *y* online, or the wavefield of the whole solution domain at that time by inputting the time *t* online. It is worth noting that this process only takes about 3 s on a single CPU workstation. PINN can accurately predict the wavefield at times much later than the training wavefield, showing satisfying generalization capability outside of the training data. Fig. 4 shows the convergence of different loss terms. The errors still show a decreasing trend, indicating that the difference between the reference solution and prediction is still possible to be smaller. Although we trained 380,000 epochs for this case, PINN has converged to a relatively accurate result

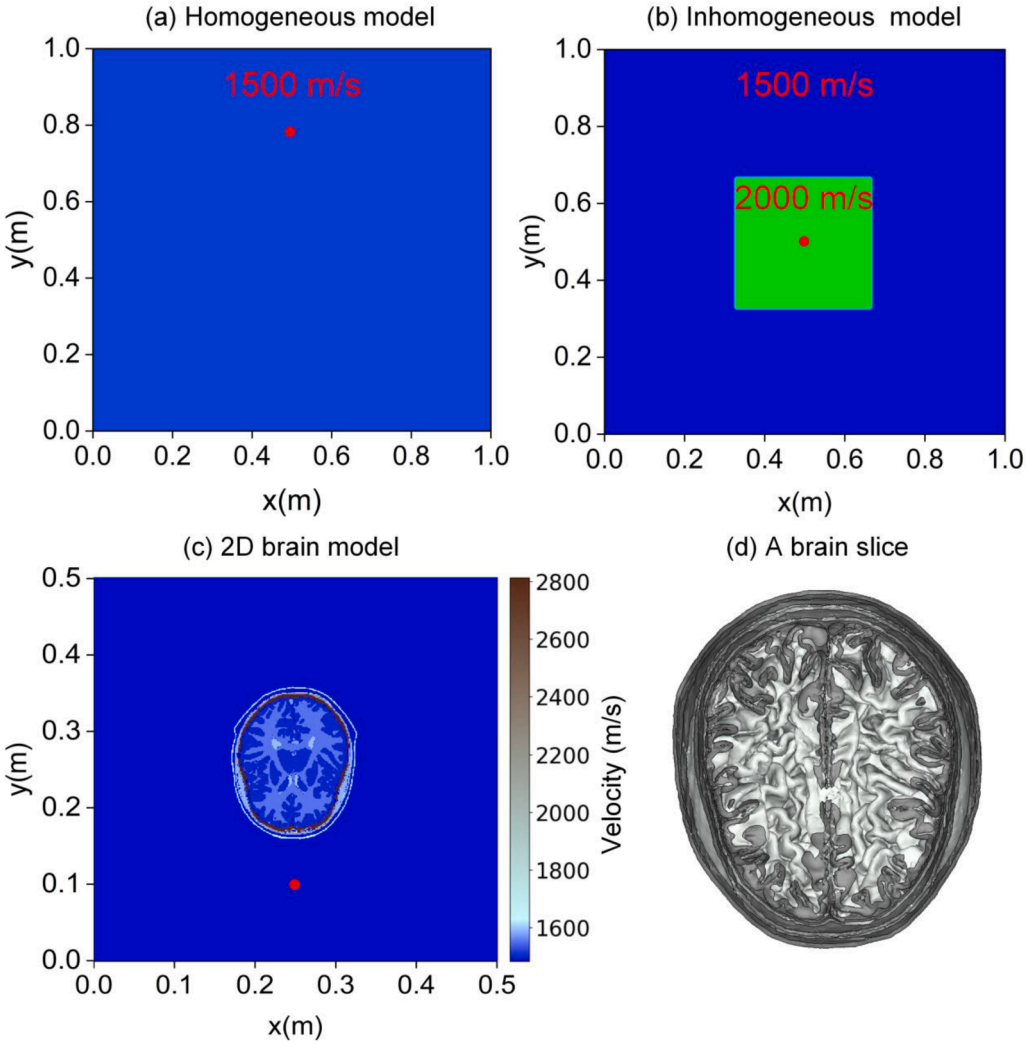


Fig. 2. Velocity models. (a) a homogeneous model; (b) an inhomogeneous model; (c) a 2D brain model transformed from (d), a slice selected from MIDA. Specific networks need to be built for each velocity model. The red circles indicate the positions of the excitation sources in the cases. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

in less than 150,000 epochs, and the training process later is used to improve the subtle errors. This is explained and illustrated in the following case of 2D brain model.

3.2. Inhomogeneous model

In this case, we explore the ability of PINNs under an inhomogeneous model with two kinds of velocity as shown in Fig. 2(b) which induces complex wavefield dynamics. In the process of propagation, the wavefield may be compressed or expanded as the velocity changes, and the reflected and transmitted waves will be formed at the interfaces of different velocities. We choose a point source with a dominant frequency of 30 kHz and a Gaussian source time function located at the center of the domain to excite the initial wavefield. Due to the increase of excitation frequency and the complexity of velocity model, we adjust the number of hidden layers to 8 and the number of neurons in each to 50. The training points of MSE_{PDE} is increased to 20,000 and we remain 6400 points for each of the early-time snapshots data. In this case, we set absorbing conditions for all the boundaries. The weights of the misfit function are $\lambda_{PDE} = 0.1$, $\lambda_{S1} = \lambda_{S2} = 1$ and $\lambda_{BC} = 0$. We simulate 0.5 *ms* wave propagation with the time step set to be 5×10^{-8} *s*. Fig. 5 shows the reference solutions run by SEM, the wavefields predicted by PINN after training and the differences between them. The wavefield solutions at 0.06 *ms*, 0.09 *ms*, 0.23 *ms* and 0.30 *ms* are compared respectively. Again, only two early-time snapshots were used for PINN training. After offline training we can obtain the time domain signal of the input coordinate (x,y), or the wavefield at the input time t. For the convenience of observation, different colormaps are used for the wavefields of the first two and the last two moments. From (a3) and (a4) of Fig. 5, we can see that although the velocity model is relatively simple, its corresponding wavefield is indeed complex, with multiple transmissions and reflections occurring at the interface where the velocity changes sharply. Still, the proposed method can reasonably capture most of the transmitted and reflected waves which are not available in the training data. It is also able to describe the phenomena of compression, expansion and amplitude attenuation of the wavefields. The maximum error is about 8% in this model. It is worth noting that only two early-time snapshots are used for training, and at these two moments the waves have not yet propagated to different velocity interfaces. That is, the constraints of the PDE play an indispensable role in the training process. However, by observing the differences in more detail, we find that the loss along the interface is a bit higher in the velocity model. One explanation may be that the discontinuity in the velocity model leads to the discontinuity of the second order gradients of the wavefields in the network, which is difficult to learn [22]. Fig. 6 shows the convergence of different loss terms.

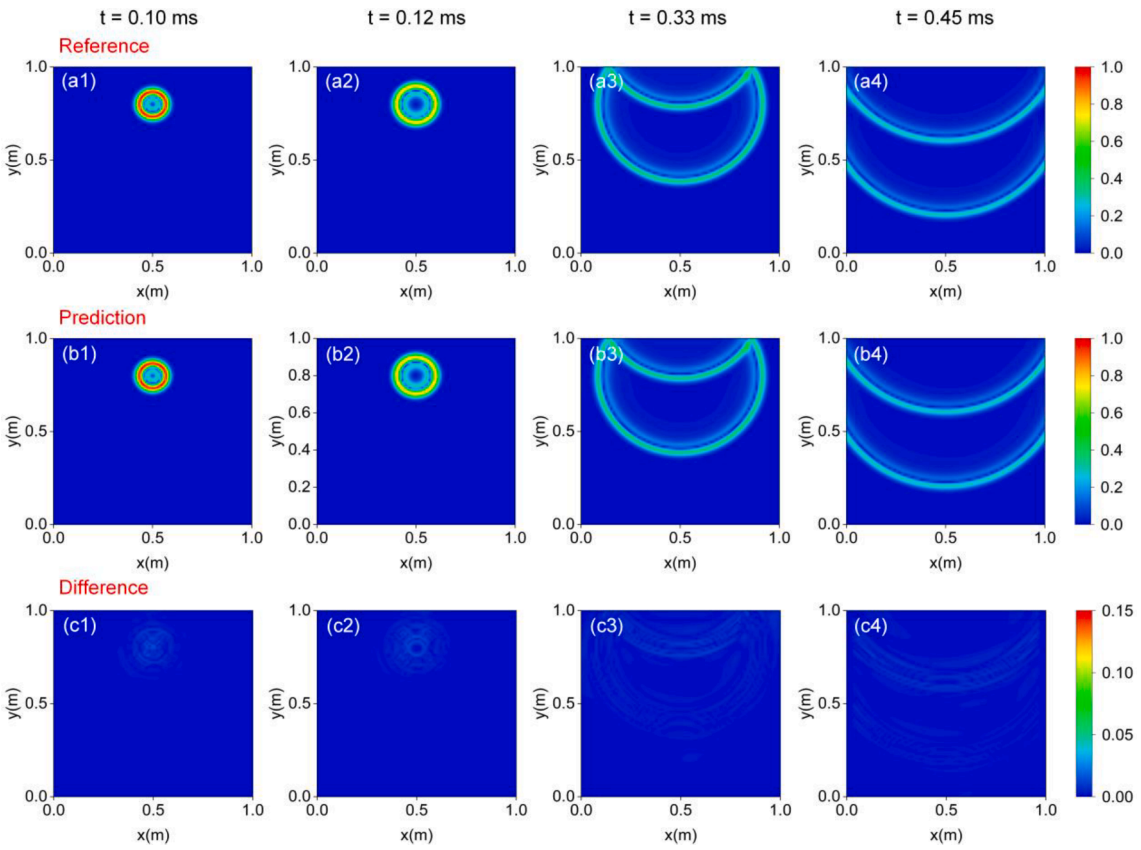


Fig. 3. Comparison of the PINN's wavefield predictions to the reference solutions using the homogeneous model. All the surfaces are absorbing boundaries except for free-surface condition for the top in this domain. (a1) to (a4) and (b1) to (b4) show the reference solutions and predictions through time. (c1) to (c4) show the differences between them. We unify the color bar ranges in order to reflect the amplitude attenuation in the process of wave propagation.

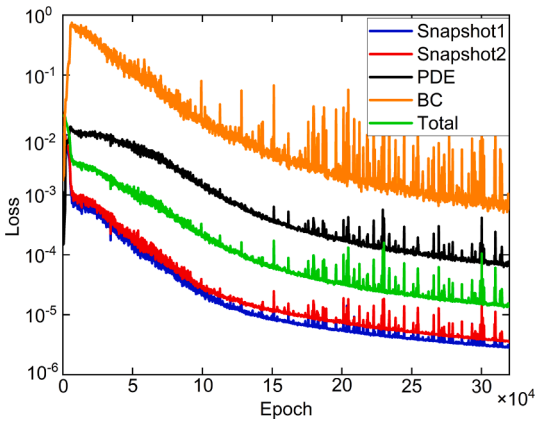


Fig. 4. The convergence of training loss for the homogeneous model.

3.3. 2D brain model

Our final case uses the 2D brain model shown in Fig. 2(c), which produces a much more complex wavefield. A Gaussian point source is located at (0.25, 0.1) with a central frequency of 100 kHz. An FNN with 10 hidden layers and 60 neurons in each layer is selected considering the complexity of the model and the high frequency. The training data sets for PDE and each of the early-time snapshots are composed of 60,000 and 10,000 data points, respectively. We impose an absorbing condition for all the surfaces of the domain. At the same time, the weights of the loss functions in Eq.(7) are adjusted to $\lambda_{PDE} = 0.01$, $\lambda_{S1} = \lambda_{S2} = 1$ and $\lambda_{BC} = 0$. The time step is set to $2 \times 10^{-8} s$, and we run 0.2 *ms* for wave propagation using the SEM. Fig. 7 shows the reference solutions, the predictions and the differences at 0.036 *ms*, 0.044 *ms*, 0.092 *ms* and 0.144 *ms*, respectively. The wavefields of Fig. 7(a1) and (a2) are used for training. Again, (a3) and (a4) are obtained by online testing. For the convenience of observation, the 2D brain velocity model is added to each result as a background, which is independent of the colormap corresponding to each result. The predictions for the first two moments match the reference solutions quite well, with a maximum error less than 5%. For the latter two moments, PINN can also predict the information of the reflection as it propagates to the skull in the brain model. However, when the wavefront propagates through the brain model, there is a loss of information of the predictions about the wavefield inside the brain. This may be due to excessive velocity differences between the skull and internal tissues and the fact that the amplitude of the information is too small to be accurately modeled by PINN. Fig. 8 shows the training loss for the 2D brain model against training step.

To understand the effect of training iterations, we recorded the predictions of PINN for the wavefield of 0.144 *ms* in this case at different epochs. Fig. 9(a1) to (a4) show the prediction results of PINN at 12,000, 20,000, 40,000 and 60,000 epochs, respectively, and (b1) to (b4) are the differences between them and the reference solution as shown in Fig. 7 (a4). Despite the complexity of the model, PINN begins to predict tiny contours of the wavefield at 12,000 epochs and has roughly converged by 20,000 epochs. At 60,000 epochs, PINN can predict the reflected information of the wavefield. The subsequent training is used to adjust the subtle wavefield information and stops at 100,000 epochs.

To further test the feasibility of PINN in predicting transcranial ultrasound wave propagation, we modified the 2D brain model to contain a clot, as shown in Fig. 10(a). We use physical characteristics for blood-infused soft tissue to establish this model [45]. The clot is defined as a circular area with a radius of 15 *mm* and a fixed sound velocity of 1600

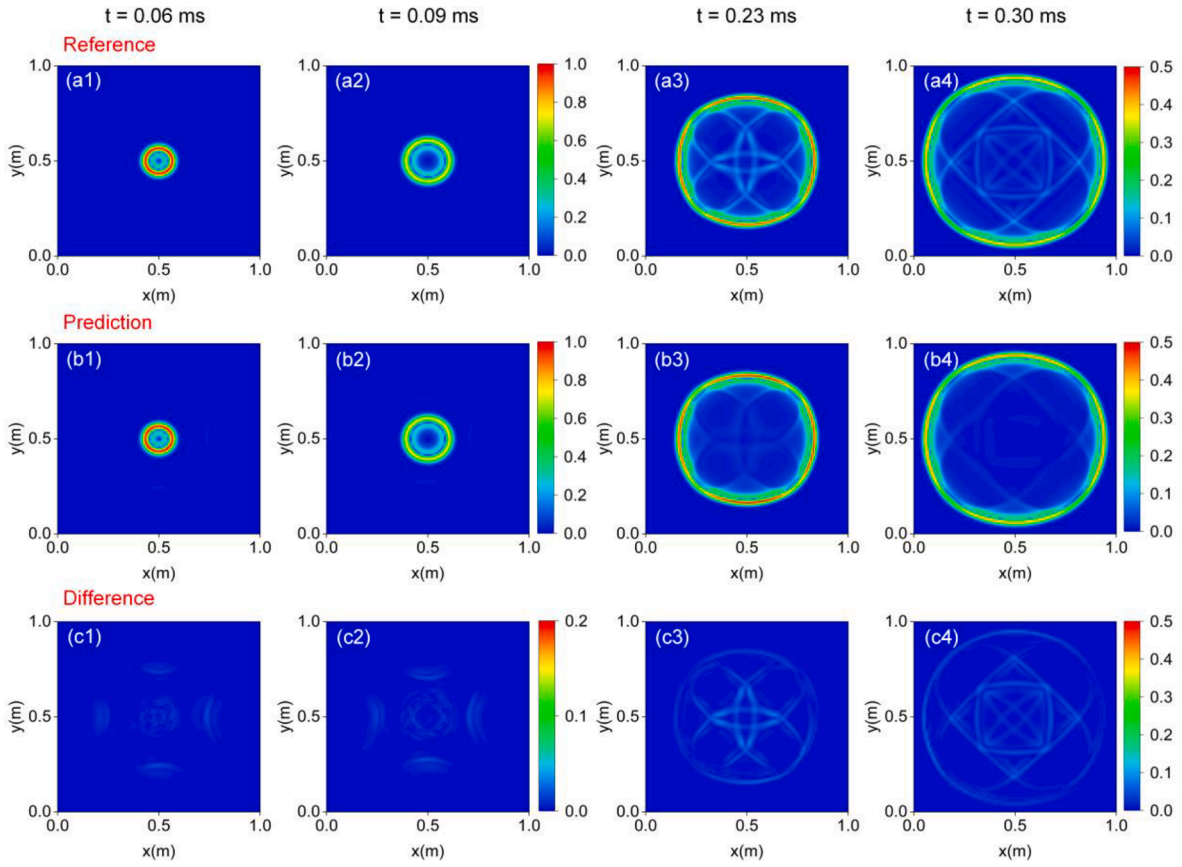


Fig. 5. Comparison of the PINN's wavefield predictions to the reference solutions using the inhomogeneous model. All the surfaces are absorbing boundaries.

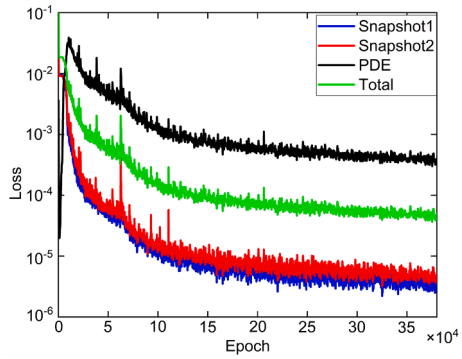


Fig. 6. The convergence of training loss for the inhomogeneous model.

m/s. Compared with the previous case, all the conditions are completely consistent except the velocity model (with or without the clot). (b) shows the reference solution obtained by the SEM at 0.144 *ms* with this model and (c) is the prediction of PINN at this moment. The difference between (b) and (c) is depicted in (d). Overall, the PINN's prediction matches the reference solution reasonably well. Similar to the case without the clot, the error is mainly reflected in the wavefront as it passes through the brain. Fig. 11 shows the differences of the wavefields between 2D brain models with and without the clot at different times obtained by the SEM. They can be regarded as the wavefields changing over time with the clot used as the source.

The match between the reference solutions and the PINN's predictions of the motion on × and y axial for the 2D brain model with the clot is shown in Fig. 12. Locations of the receiving sensors are shown as the green triangles in Fig. 10(a). They are evenly distributed between (0.40,0.10) and (0.40,0.30). We can see there are two peaks in the × axial motion, of which the front one represents the signal propagating forward and the other one represents the reflected signal. For some of the reflected signals, there are some discrepancies between the reference solutions and predictions. However, for the fluctuating trend of the time domain signal, the solution of PINN closely matches the reference, which proves its ability of calculating the wavefield. Fig. 13 shows the comparison of the spectrum on × and y axial at (0.40,0.30). The spectrum of the prediction is basically consistent with the reference, with the center frequency near 100 kHz.

4. Discussion

4.1. Advantages of this method

We have demonstrated that PINNs can solve the problem of wave propagation for transcranial ultrasound by using several increasingly complex spatial velocity models. The complexity of the network (such as the number of hidden layers and neurons) increases with the complexity of the problem (such as frequency and velocity models) [46]. For the same velocity model, modeling for higher frequencies requires more PDE training points to capture finer features, such as multiple reflected waves. PINNs are mesh-free and consequently can be extended to arbitrarily complex computational boundaries with relative ease. Impressively, they can predict subtle physical phenomena such as transmitted and reflected waves at different velocity interfaces, wavefield compression and expansion, and amplitude attenuation of wavefields at much later times than the training data, which is a promising approach among existing deep learning approaches. Compared with traditional methods, this deep learning-based method improves online prediction by transferring excessive computational load to offline by training a NN model. After the network training is completed, we use PINN and SEM to generate ten wavefields successively on the same device (a single CPU

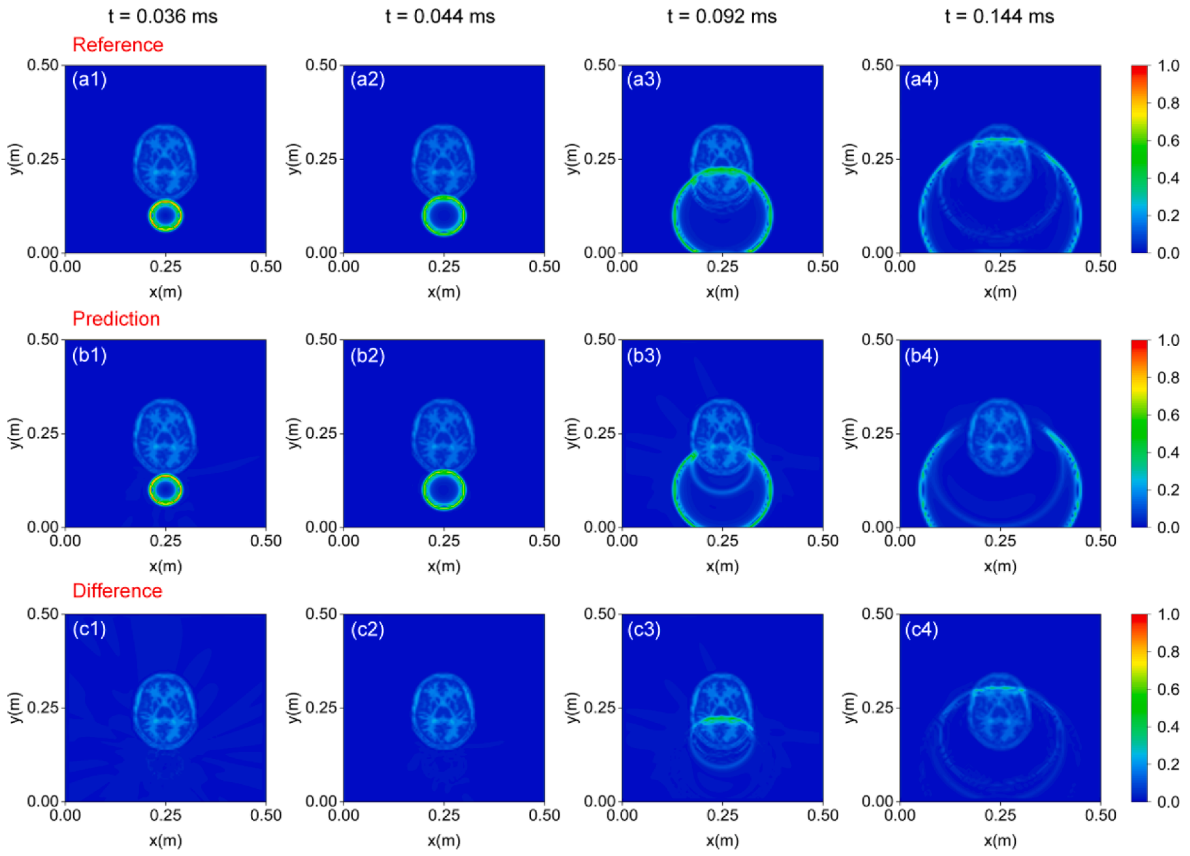


Fig. 7. Comparison of the PINN's wavefield predictions to the reference solutions using the 2D brain model. All the surfaces are absorbing boundaries.

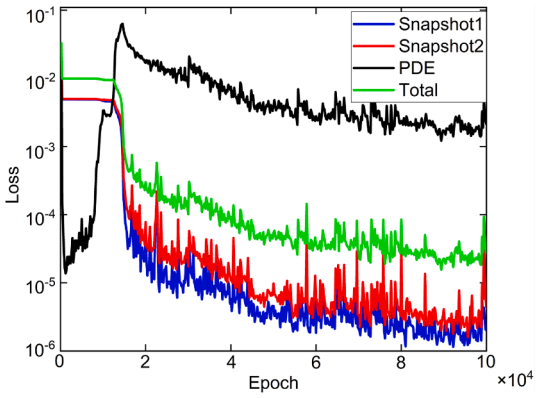


Fig. 8. The convergence of training loss for the 2D brain model.

workstation) to compare the computation time. It should be noted that we set the same time step in advance. Table 1 shows the comparison results for the three models. For all the three models, the computation time of PINN is smaller than that of SEM. In addition, it is also notable that when the model becomes complex, the computation time of PINN is not greatly affected, while the required time of SEM changes significantly. Meanwhile, PINN can directly obtain the wavefield at the later time, while SEM must iterate from the initial time to the later time. These clearly demonstrate the superiority of PINN in computation.

In addition, since the method is meshless, it avoids the dispersion problem that traditional mesh-based methods typically encounter when the mesh is coarse. PINNs could also be used for source inversion, as it is readily accessible to obtain the gradients of the input terms and it is fully flexible in the selection of measurement positions for the observed wavefield. Another benefit of this approach is that we can quantify how well the network satisfies the corresponding physical constraints through each loss term, which helps to evaluate the accuracy and reliability of the network and improve its performance [22].

4.2. Limitations and solutions

In this work we find that modeling at the interface with high velocity contrast is the main cause of errors because the second-order derivatives of the wavefields at these locations change discretely, which is a challenge for PINNs to learn. In fact, this problem is also tricky for traditional PDE solvers, which usually take an oversampling approach to guarantee the accuracy of the solutions [39]. For PINN, this situation can be improved to a certain extent by adjusting the training parameters such as network depth and number of iterations. It is also a good solution to choose a more general wave equation. Another related issue is that PINNs have difficulty in simulating reflected waves of relatively low amplitudes in 2D brain model. Future work could explore these issues, for example, by using specific networks for different velocity regions to reduce the impact of network discontinuities, by trying different activation functions [47], by adding physical constraint terms to make the loss function more relevant to the practical problems, or by adopting smarter sampling schemes [48]. Admittedly, we use data from SEM as the training set, so the accuracy of PINN's results will not be better than that of SEM. However, it is undeniable that PINN improves the computational efficiency by transferring computational load to offline training, and has the advantages of no dispersion, easy dealing with different BCs, and not requiring a large amount of label data, which are the aspects we would like to emphasize. The proposed method may bring new enlightenment to the field of transcranial ultrasound.

In this paper, a signal with a dominant frequency of 100 kHz is chosen for the simulation of the transcranial wavefield. This frequency is relatively low. In fact, if we choose a higher frequency (e.g., 0.5 MHz),

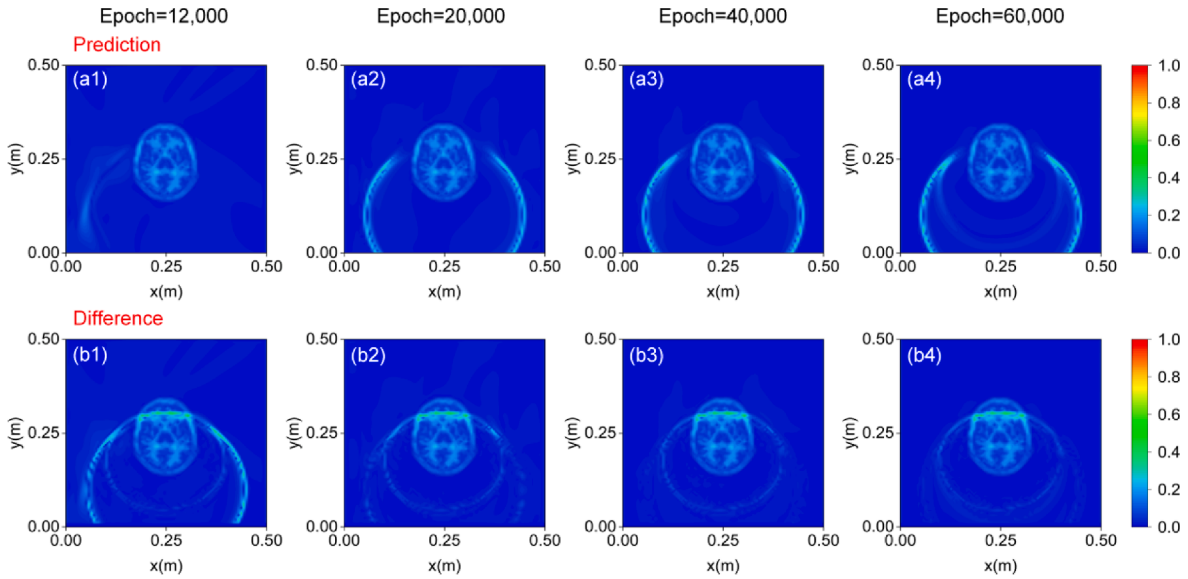


Fig. 9. Comparison of the PINN's wavefield predictions to the reference solution at different epochs for the 2D brain model.

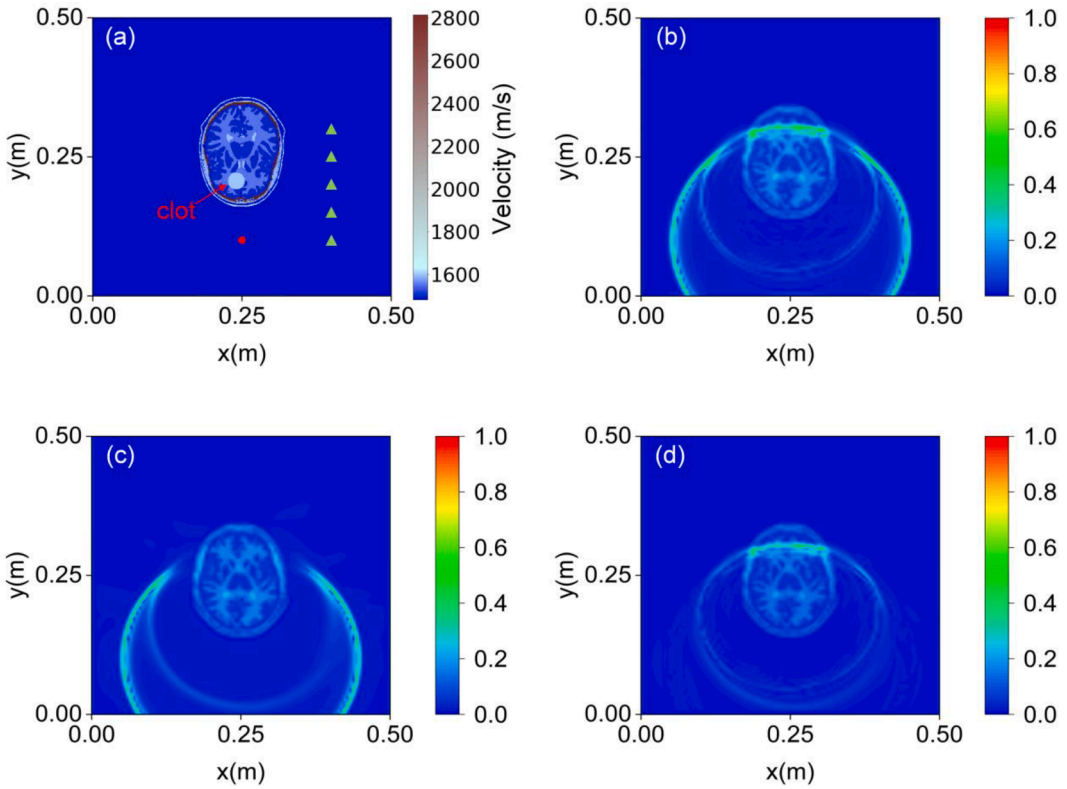


Fig. 10. Comparison of the PINN's wavefield prediction to the reference solution using the 2D brain model with a clot. (a) 2D brain model with a clot (the green triangles represent the receiving sensors); (b) the reference solution obtained by the SEM; (c) the prediction of PINN; (d) the difference between (b) and (c). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

SEM needs to divide the grid several times more to obtain a wavefield without dispersion. When using the snapshot data under such condition for PINN training, more sampling points are also required to ensure the accuracy of the predicting results, which is not allowed by the current computational power. It seems that PINNs require a higher computational cost to train than numerical solutions in some applications. When the velocity model or the position of the excitation source changes, PINN needs to be retrained. However, on the one hand, traditional methods also face this problem; on the other hand, PINN can improve the generalization capability by taking the coordinates of the excitation source as the input of the network, and can improve the computational efficiency by applying the trained network to the model with little difference through transfer learning, which has been investigated. In addition, we have been already working on the prediction of the transcranial wavefield in frequency domain by embedding the Helmholtz equation into the loss function. The prediction in frequency domain will also save computational resources compared to time domain. This method has the potential to pave a new way for

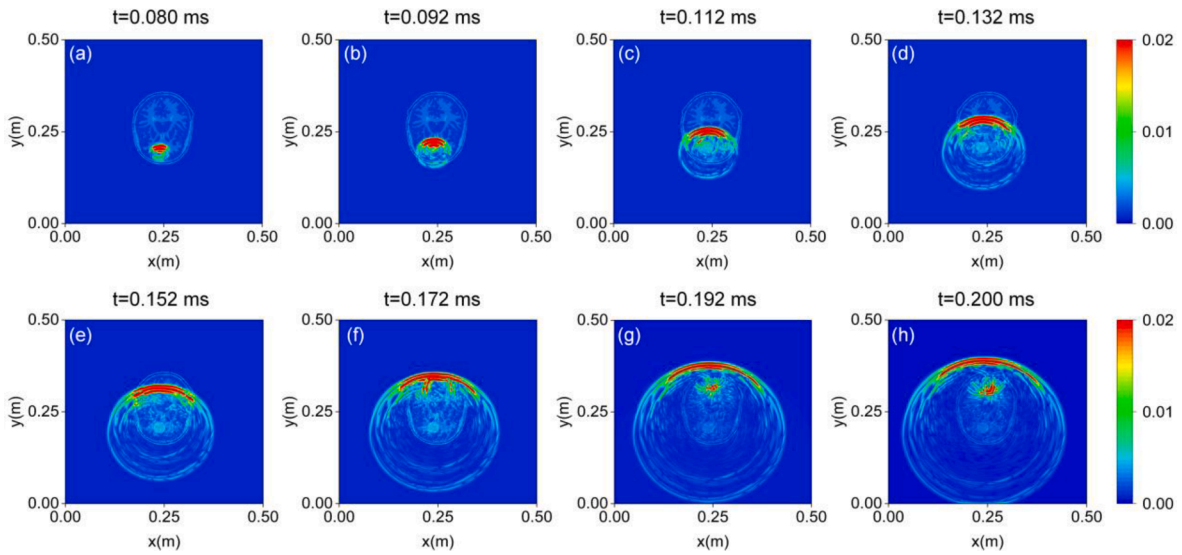


Fig. 11. The differences of the wavefields between 2D brain model with and without a clot at different times obtained by the SEM.

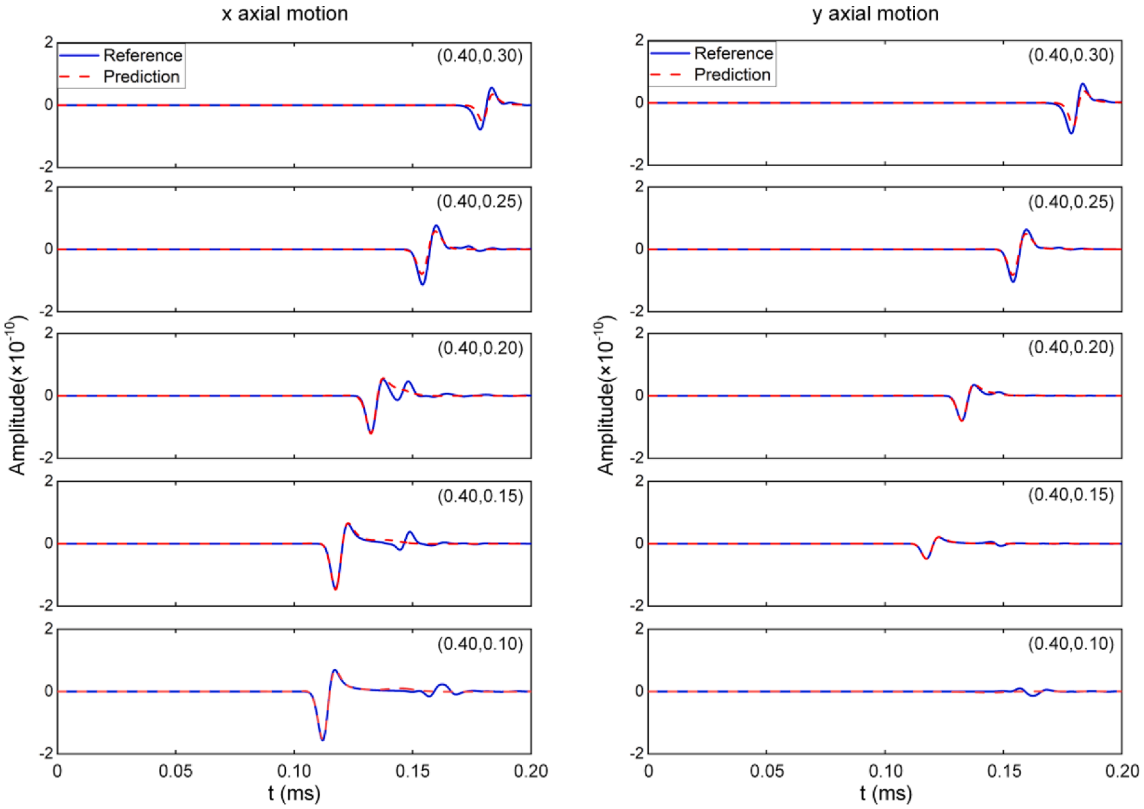


Fig. 12. The comparison between the reference solutions (blue line) and the PINN's predictions (red dashed line) of the motion on × and y axial for the 2D brain model with a clot. Locations of the receiving sensors are shown as green triangles in Fig. 10(a). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

applications in other fields. This includes applications in more complex physics, like viscoelastic anisotropic medium and fluid-saturated porous medium, where the traditional wave equation solvers become impractical, especially in the case of large model sizes. Deep learning has the ability to learn nonlinear relationships in a given problem by identifying patterns and making accurate judgments and predictions about them, making it one of the best methods for dealing with large data and large problems. For complex physical problems, the loss function may be changed with the addition of more constraint terms and the adjustment of the weights, but the network shares the same training mechanism and does not need to solve the inverse of large matrices.

4.3. Future research directions

Finally, our approach can be easily extended to 3D and more complex wave equations to model the transcranial ultrasound propagation more accurately. 3D extension is trivial in the network design, requiring only one additional dimension of the input tensor. Furthermore,

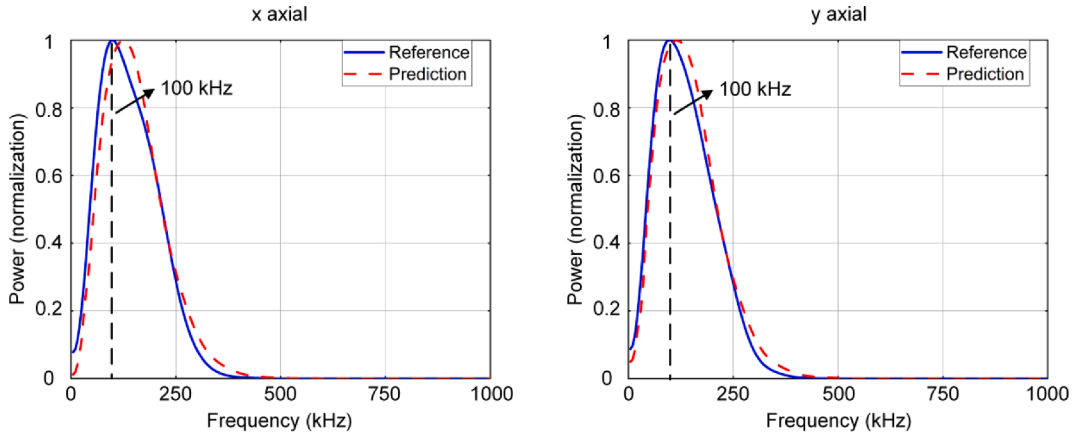


Fig. 13. The comparison between the reference solutions (blue line) and the PINN's predictions (red dashed line) of the spectrum on × and y axial at (0.40,0.30). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Table 1
The computation time of PINN and SEM to obtain ten wavefields (Unit: seconds).

Model	SEM	PINN
Homogeneous model	12.91	10.25
Inhomogeneous model	29.20	14.44
2D brain model	41.40	16.90

generating 3D training data may not be too challenging in practice, as the proposed approach only requires a few early-time snapshots of reference solutions to train the network. For more complex situations, such as the elastic wave equation in more complex medium, we need to modify the loss function term in Eq.(7). The performance of the proposed method for transcranial ultrasound wave propagation indicates that PINNs may also do well for other complex physical systems. Our future work will investigate these directions.

5. Conclusion

In this paper, we solve the problem of transcranial ultrasound wave propagation by applying PINNs to 2D acoustic wave equation. We use a fully connected FNN, which takes spatial coordinates as inputs and the wavefield solution as output to the network. The performance of PINNs is verified by a homogeneous model, an inhomogeneous model and a 2D brain model with and without a clot. The results show that the proposed method has an impressive generalization capability and can predict the wavefield solution much later in time given only two early-time snapshots for training. Future work includes its application to 3D brain imaging through 3D physical equations and the consideration of the viscoelastic anisotropic medium or fluid-saturated porous medium, which is closer to the real situation. More work needs to be done to extend it for more general model configurations (e.g., arbitrary velocity models, or velocity models with limited variations) before it can be used for imaging.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

Acknowledgements

This work was supported in part by the National Science Foundation of China under contract number 61773283, in part by National Key R&D Program of China under contract number 2018YFC0808600.

REFERENCES

[1] D. Antipova, L. Eadie, S. Makin, H. Shannon, P. Wilson, A. Macaden, The use of transcranial ultrasound and clinical assessment to diagnose ischaemic stroke due to large vessel occlusion in remote and rural areas, PLoS One 15 (10) (2020) e0239653.
[2] A.E. Roher, Z. Garami, S.L. Tyas, C.L. Maarouf, T.A. Kokjohn, M. Belohlavek, L. J. Vedders, D. Connor, M.N. Sabbagh, T.G. Beach, M.R. Emmerling, Transcranial Doppler ultrasound blood flow velocity and pulsatility index as systemic indicators for Alzheimer's disease, Alzheimers Dement. 7 (4) (2011) 445–455.
[3] A. Gaenslen, B. Unmuth, J. Godau, I. Liepelt, A. Di Santo, K.J. Schweitzer, T. Gasser, H.-J. Machulla, M. Reimold, K. Marek, D. Berg, The specificity and sensitivity of transcranial ultrasound in the differential diagnosis of Parkinson's disease: a prospective blinded study, Lancet Neurol. 7 (5) (2008) 417–424.
[4] M. Mozaffarzadeh, E. Verschuur, M.D. Verweij, V. Daeichin, N. De Jong, G. Renaud, Refraction-Corrected Transcranial Ultrasound Imaging Through the Human Temporal Window Using a Single Probe, IEEE Trans. Ultrason. Ferroelectr. Freq. Control. 69 (4) (2022) 1191–1203.
[5] W. Lee, H.-C. Kim, Y. Jung, Y.A. Chung, I.-U. Song, J.-H. Lee, S.-S. Yoo, Transcranial focused ultrasound stimulation of human primary visual cortex, Sci Rep. 6 (1) (2016) 1–12.
[6] M. Almekkawy, J. Chen, M.D. Ellis, D. Haemmerich, D.R. Holmes, C.A. Linte, D. Panescu, J. Pearce, P. Prakash, V. Zderic, Therapeutic systems and technologies: State-of-the-art applications, opportunities, and challenges, IEEE Rev. Biomed. Eng. 13 (2019) 325–339.
[7] J. Gu, Y. Jing, Modeling of wave propagation for medical ultrasound: a review, IEEE Trans. Ultrason. Ferroelectr. Freq. Control. 62 (11) (2015) 1979–1992.
[8] H. Wang, J. Li, L. Wang, L. Liang, Z. Zeng, Y. Liu, On acoustic fields of complex scatters based on physics-informed neural networks, Ultrasonics 106872 (2022).
[9] J. Tong, M. Lin, X. Wang, J. Li, J. Ren, L. Liang, Y. Liu, Deep learning inversion with supervision: A rapid and cascaded imaging technique, Ultrasonics 122 (2022), 106686.
[10] B. Vafaeian, M. El-Rich, T. El-Bialy, S. Adeeb, The finite element method for micro-scale modeling of ultrasound propagation in cancellous bone, Ultrasonics 54 (6) (2014) 1663–1676.
[11] K. Nakahata, H. Sugahara, M. Barth, B. Köhler, F. Schubert, Three dimensional image-based simulation of ultrasonic wave propagation in polycrystalline metal using phase-field modeling, Ultrasonics 67 (2016) 18–29.
[12] Z. Guo, G. Egbert, H. Dong, W. Wei, Modular finite volume approach for 3D magnetotelluric modeling of the earth medium with general anisotropy, Phys. Earth Planet. Inter. 309 (2020), 106585.
[13] C. Song, T. Alkhalifah, U.B. Waheed, Solving the frequency-domain acoustic VTI wave equation using physics-informed neural networks, Geophys. J. Int. 225 (2) (2021) 846–859.
[14] S. Alkhadhr, X. Liu, M. Almekkawy, Modeling of the Forward Wave Propagation Using Physics-Informed Neural Networks, in, IEEE International Ultrasonics Symposium (IUS). 2021 (2021) 1–4.
[15] S. Panghal, M. Kumar, Optimization free neural network approach for solving ordinary and partial differential equations, Eng. Comput. 37 (4) (2020) 2989–3002.
[16] T.W. Hughes, I.A. Williamson, M. Minkov, S. Fan, Wave physics as an analog recurrent neural network, Sci. Adv. 5 (12) (2019) eaay6946.

[17] B. Moseley, T. Nissen-Meyer, A. Markham, Deep learning for fast simulation of seismic waves in complex media, Solid Earth 11 (4) (2020) 1527–1549.
[18] T. Gantala, K. Balasubramaniam, DPAI: A Data-driven simulation-assisted-Physics learned AI model for transient ultrasonic wave propagation, Ultrasonics 121 (2022), 106671.
[19] S. Ahmed, U. Kamal, M.K. Hasan, DSWE-Net: A deep learning approach for shear wave elastography and lesion segmentation using single push acoustic radiation force, Ultrasonics 110 (2021), 106283.
[20] A. Stanziola, S.R. Arridge, B.T. Cox, B.E. Treeby, A Helmholtz equation solver using unsupervised learning: Application to transcranial ultrasound, J. Comput. Phys. 441 (2021), 110430.
[21] M. Raissi, P. Perdikaris, G.E. Karniadakis, Physics-informed neural networks: A deep learning framework for solving forward and inverse problems involving, J. Comput. Phys. 378 (2019) 686–707.
[22] B. Moseley, A. Markham, T. Nissen-Meyer, Solving the wave equation with physics-informed deep learning, arXiv preprint arXiv:2006.11894, 2020.
[23] A.G. Baydin, B.A. Pearlmutter, A.A. Radul, J.M. Siskind, Automatic differentiation in machine learning: a survey, J. Mach. Learn. Res. 18 (2018) 1–43.
[24] D. Voytan, M. K. Sen, Wave propagation with physics informed neural networks, in: SEG International Exposition and Annual Meeting, 2020.
[25] S. Karimpouli, P. Tahmasebi, Physics informed machine learning: Seismic wave equation, Geosci. Front. 11 (6) (2020) 1993–2001.
[26] Z. Mao, A.D. Jagtap, G.E. Karniadakis, Physics-informed neural networks for high-speed flows, Comput. Meth. Appl. Mech. Eng. 360 (2020), 112789.
[27] M. Rautela, J. Senthilnath, J. Moll, S. Gopalakrishnan, Combined two-level damage identification strategy using ultrasonic guided waves and physical knowledge assisted machine learning, Ultrasonics 115 (2021), 106451.
[28] R. L. van Herten, A. Chiribiri, M. Breeuwer, M. Veta, C. M Scannell, Physics-informed neural networks for myocardial perfusion MRI quantification, arXiv preprint arXiv:2011.12844, 2020.
[29] Z. Chen, Y. Liu, H. Sun, Physics-informed learning of governing equations from scarce data, Nat. Commun. 12 (1) (2021) 6136.
[30] M. Raissi, A. Yazdani, G.E. Karniadakis, Hidden fluid mechanics: Learning velocity and pressure fields from flow visualizations, Science 367 (6481) (2020) 1026–1030.
[31] L. Lu, X. Meng, Z. Mao, G.E. Karniadakis, DeepXDE: A deep learning library for solving differential equations, SIAM Rev. 63 (1) (2021) 208–228.
[32] A. Pulkkinen, B. Werner, E. Martin, K. Hynynen, Numerical simulations of clinical focused ultrasound functional neurosurgery, Phys. Med. Biol. 59 (7) (2014) 1679–1700.
[33] P.J. White, G.T. Clement, K. Hynynen, Longitudinal and shear mode ultrasound propagation in human skull bone, Ultrasound Med. Biol. 32 (7) (2006) 1085–1096.
[34] J. Robertson, E. Martin, B.T. Cox, B.E. Treeby, Sensitivity of simulated transcranial ultrasound fields to acoustic medium property maps, Phys. Med. Biol. 62 (7) (2017) 2559–2580.
[35] F. Marquet, M. Pernot, J.-F. Aubry, G. Montaldo, L. Marsac, M. Tanter, M. Fink, Non-invasive transcranial ultrasound therapy based on a 3D CT scan: protocol validation and in vitro results, Phys. Med. Biol. 54 (9) (2009) 2597–2613.
[36] P.B. Rosnitskiy, P.V. Yuldashev, O.A. Sapozhnikov, L.R. Gavrilov, V.A. Khokhlova, Simulation of nonlinear trans-skull focusing and formation of shocks in brain using a fully populated ultrasound array with aberration correction, J. Acoust. Soc. Am. 146 (3) (2019) 1786–1798.
[37] J.-F. Aubry, M. Tanter, M. Pernot, J.-L. Thomas, M. Fink, Experimental demonstration of noninvasive transskull adaptive focusing based on prior computed tomography scans, J. Acoust. Soc. Am. 113 (1) (2003) 84–93.
[38] B.E. Treeby, J. Jaros, A.P. Rendell, B.T. Cox, Modeling nonlinear ultrasound propagation in heterogeneous media with power law absorption using a k-space pseudospectral method, J. Acoust. Soc. Am. 131 (6) (2012) 4324–4336.
[39] H. Igel, Computational seismology: a practical introduction, Oxford University Press, 2017.
[40] D. P. Kingma, J. Ba, Adam: A method for stochastic optimization, arXiv preprint arXiv:1412.6980, 2015.
[41] Q. He, D. Barajas-Solano, G. Tartakovsky, A.M. Tartakovsky, Physics-informed neural networks for multiphysics data assimilation with application to subsurface transport, Adv. Water Resour. 141 (2020).
[42] B.E. Treeby, B.T. Cox, Modeling power law absorption and dispersion in viscoelastic solids using a split-field and the fractional Laplacian, J. Acoust. Soc. Am. 136 (4) (2014) 1499–1510.
[43] M.I. Iacono, E. Neufeld, E. Akinnagbe, K. Bower, J. Wolf, I. Vogiatzis Oikonomidis, D. Sharma, B. Lloyd, B.J. Wilm, M. Wyss, K.P. Pruessmann, A. Jakab, N. Makris, E. D. Cohen, N. Kuster, W. Kainz, L.M. Angelone, MIDA: A multimodal imaging-based detailed anatomical model of the human head and neck, PLoS One 10 (4) (2015) e0124126.
[44] L. Guasch, Ò. Calderón Agudo, M.-X. Tang, P. Nachev, M. Warner, Full-waveform inversion imaging of the human brain, npj Digit, Med. 3 (1) (2020) 1–12.
[45] C.-C. Huang, Y.-H. Lin, T.-Y. Liu, P.-Y. Lee, S.-H. Wang, Review: Study of the blood coagulation by ultrasound, J. Med. Biol. Eng. 31 (2) (2011) 79–86.
[46] T. Alkhalifah, C. Song, U. bin Waheed, Q. Hao, Wavefield solutions from machine learned functions constrained by the Helmholtz equation, Artif Intell Rev. 2 (2021) 11-19.
[47] V. Sitzmann, J. Martel, A. Bergman, D. Lindell, G. Wetzstein, Implicit neural representations with periodic activation functions, Adv. Neural Inf. Process. Syst. 33 (2020) 7462–7473.
[48] C. Anitescu, E. Atroshchenko, N. Alajlan, T. Rabczuk, Artificial neural network methods for the solution of second order boundary value problems, Comput. Mater. Contin. 59 (1) (2019) 345–359.
[49] X. Wang, J. Li, D. Wang, et al., Sparse ultrasonic guided wave imaging with compressive sensing and deep learning[J], Mech. Syst. Signal Process. 178 (2022) 109346.
[50] J. Lin, J. Li, C. Jiang, et al., Theoretical and experimental investigation of circumferential guided waves in orthotropic annuli[J], Ultrasonics 123 (2022) 106715.